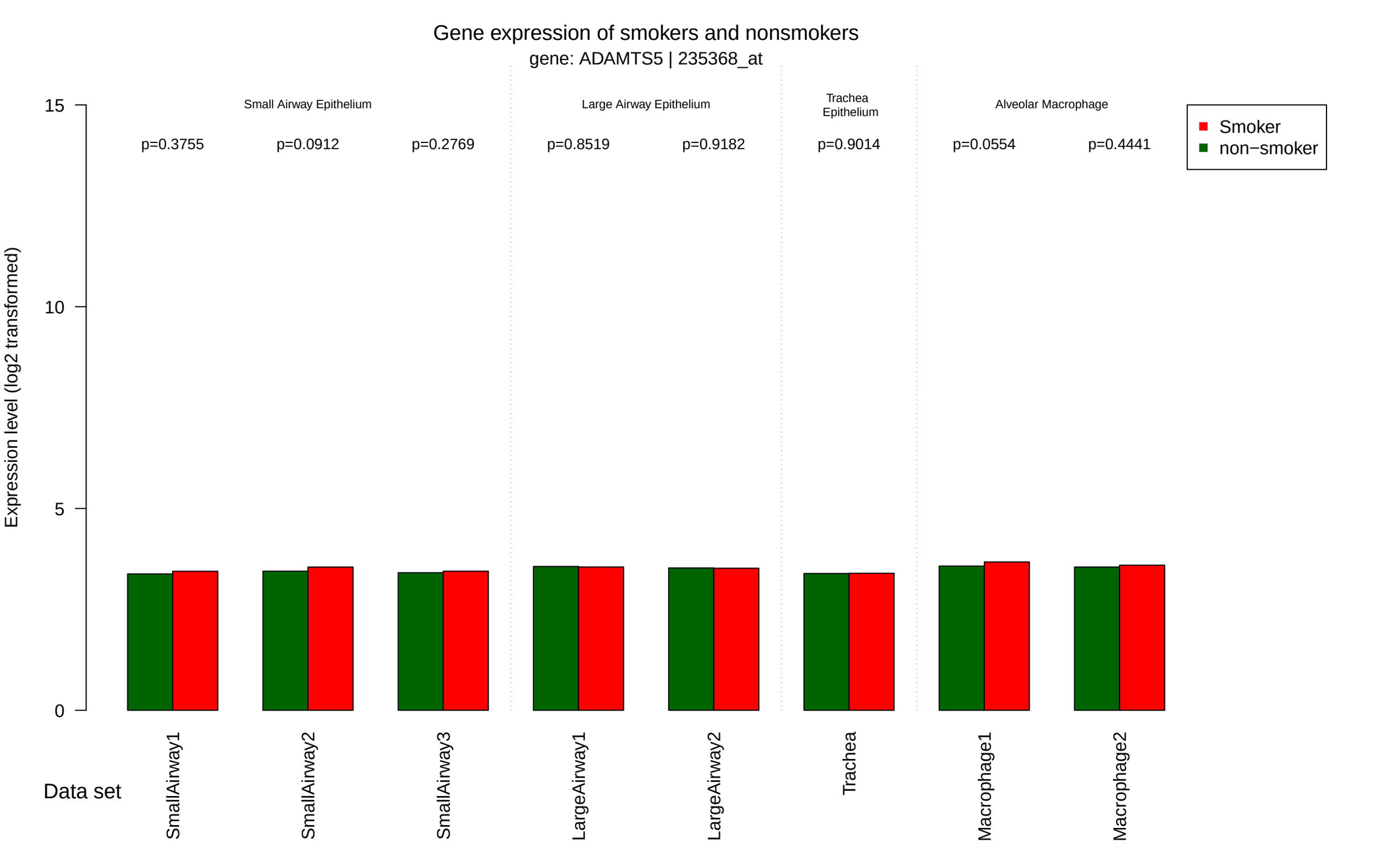
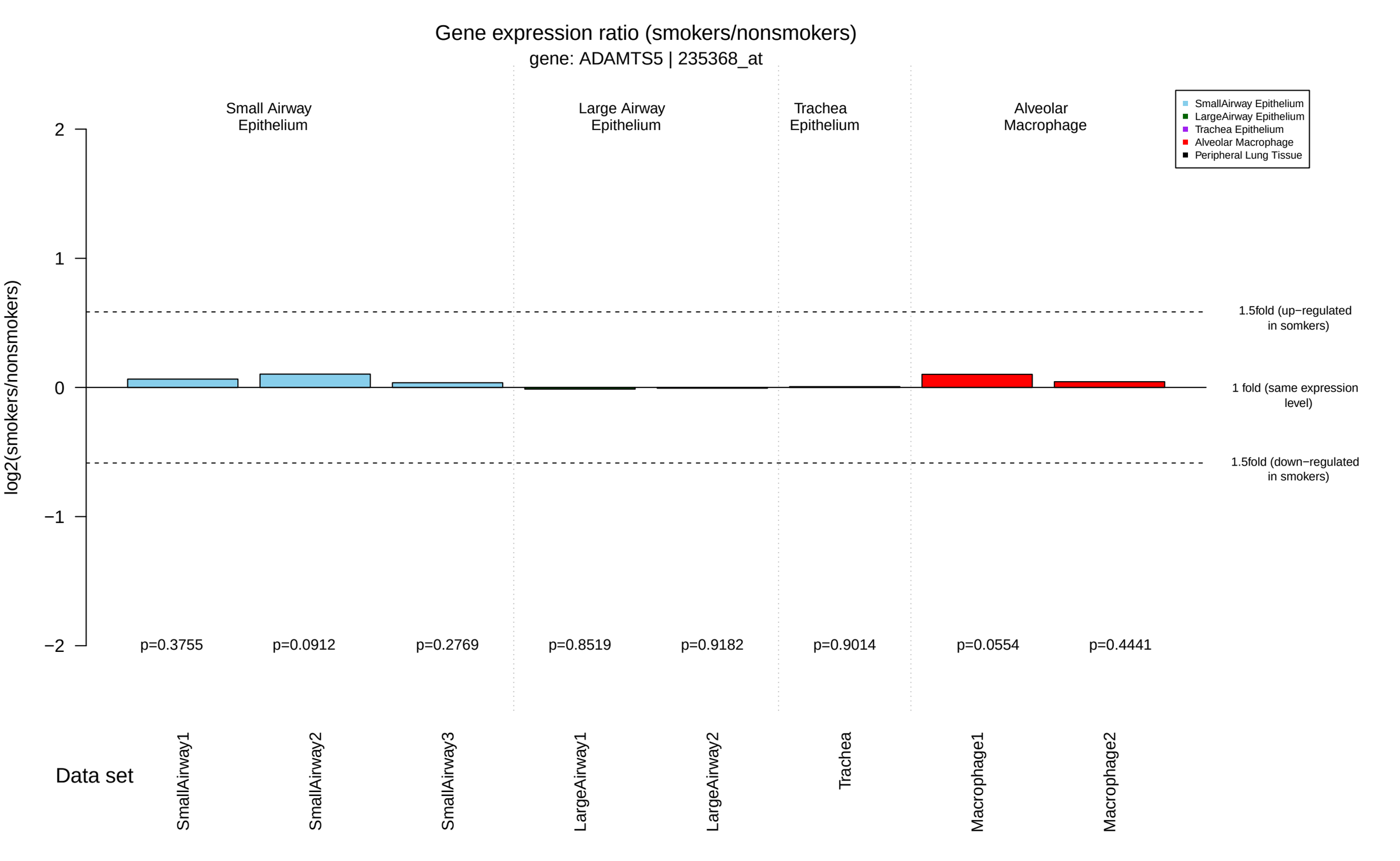
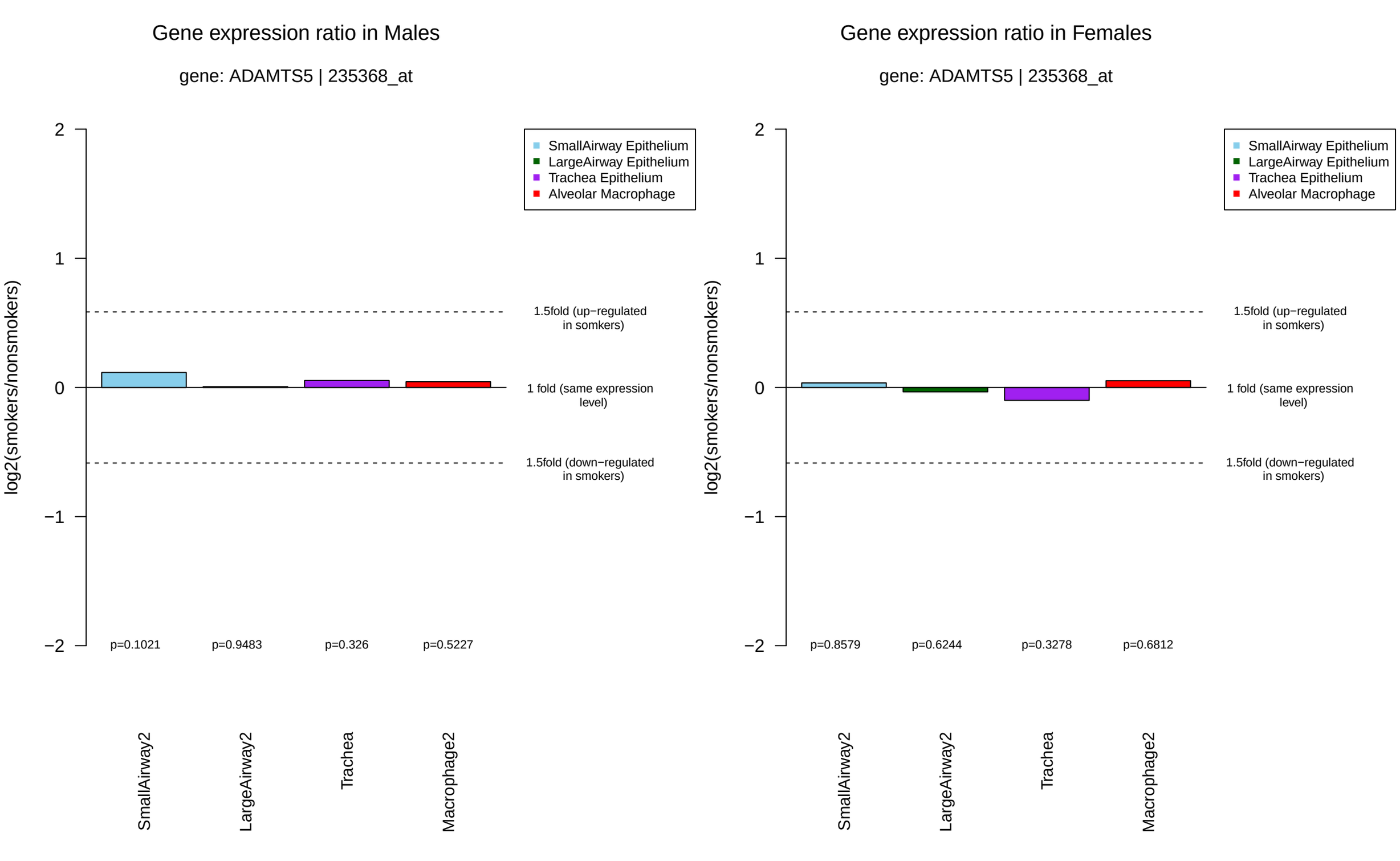
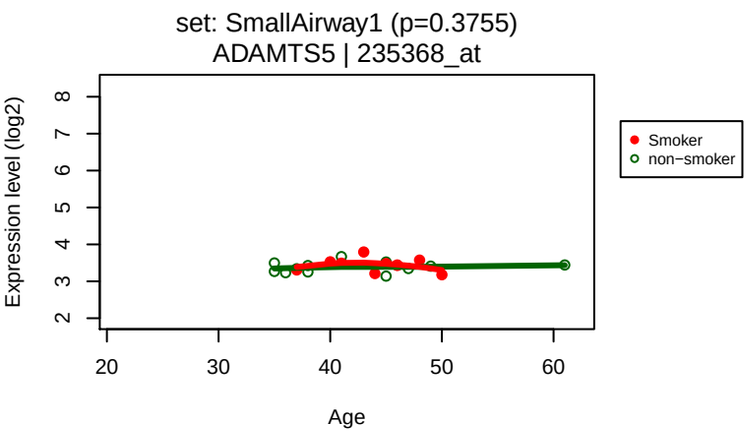
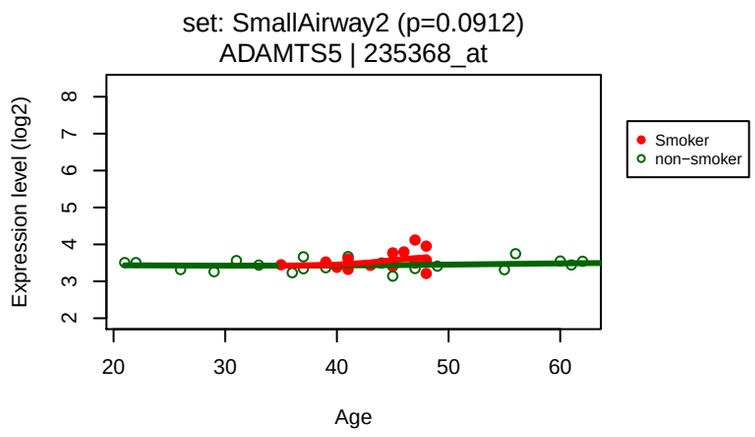
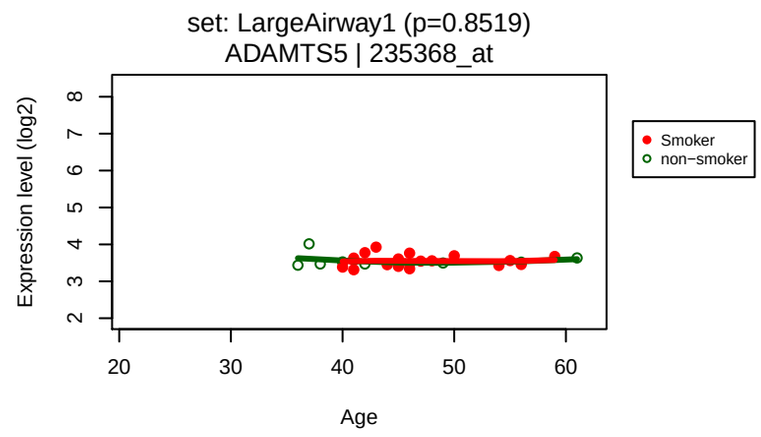

Gene expression of smokers and nonsmokers
gene: ADAMTS5 | 235368_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3755
p=0.0912
p=0.2769
p=0.8519
p=0.9182
p=0.9014
p=0.0554
p=0.4441
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

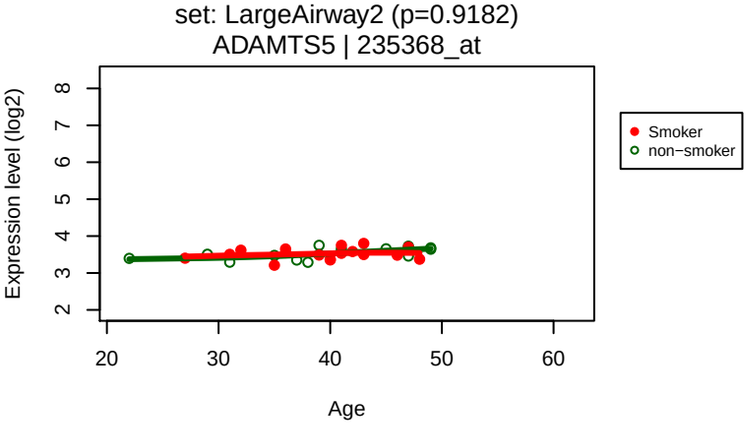

Gene expression ratio (smokers/nonsmokers)
gene: ADAMTS5 | 235368_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3755
p=0.0912
p=0.2769
p=0.8519
p=0.9182
p=0.9014
p=0.0554
p=0.4441
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

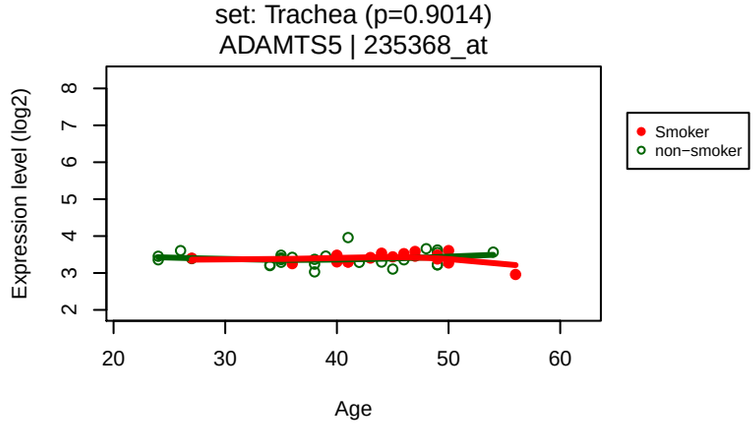

Gene expression ratio in Males
gene: ADAMTS5 | 235368_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1021
p=0.9483
p=0.326
p=0.5227
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADAMTS5 | 235368_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8579
p=0.6244
p=0.3278
p=0.6812
SmallAirway2
LargeAirway2
Trachea
Macrophage2

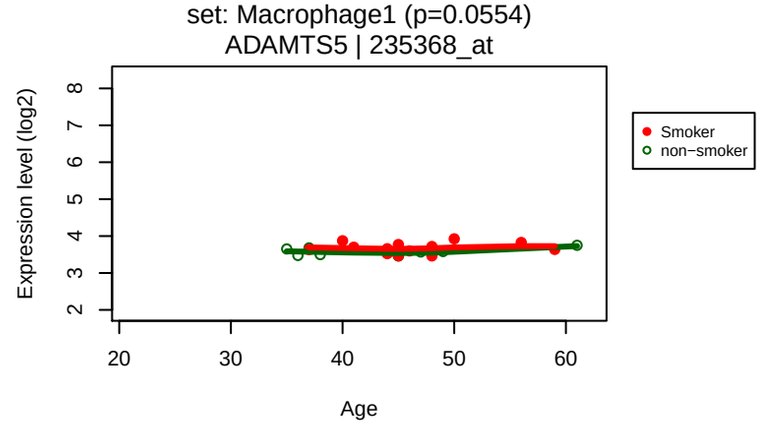

set: SmallAirway1 (p=0.3755)
ADAMTS5 | 235368_at
Expression level (log2)
Age
Smoker
non−smoker

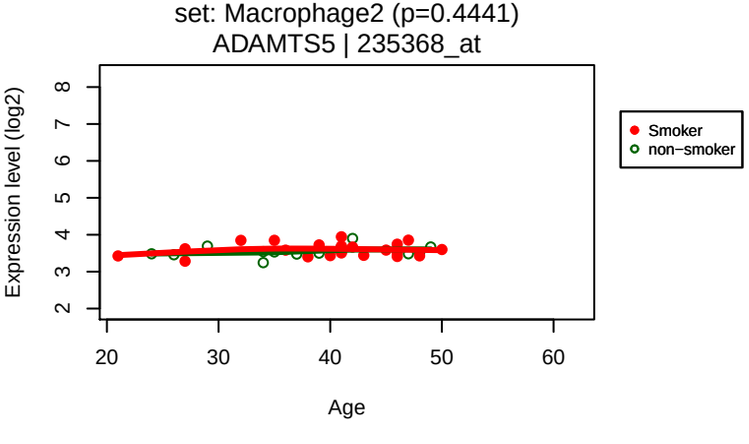

set: SmallAirway2 (p=0.0912)
ADAMTS5 | 235368_at
Expression level (log2)
Age
Smoker
non−smoker

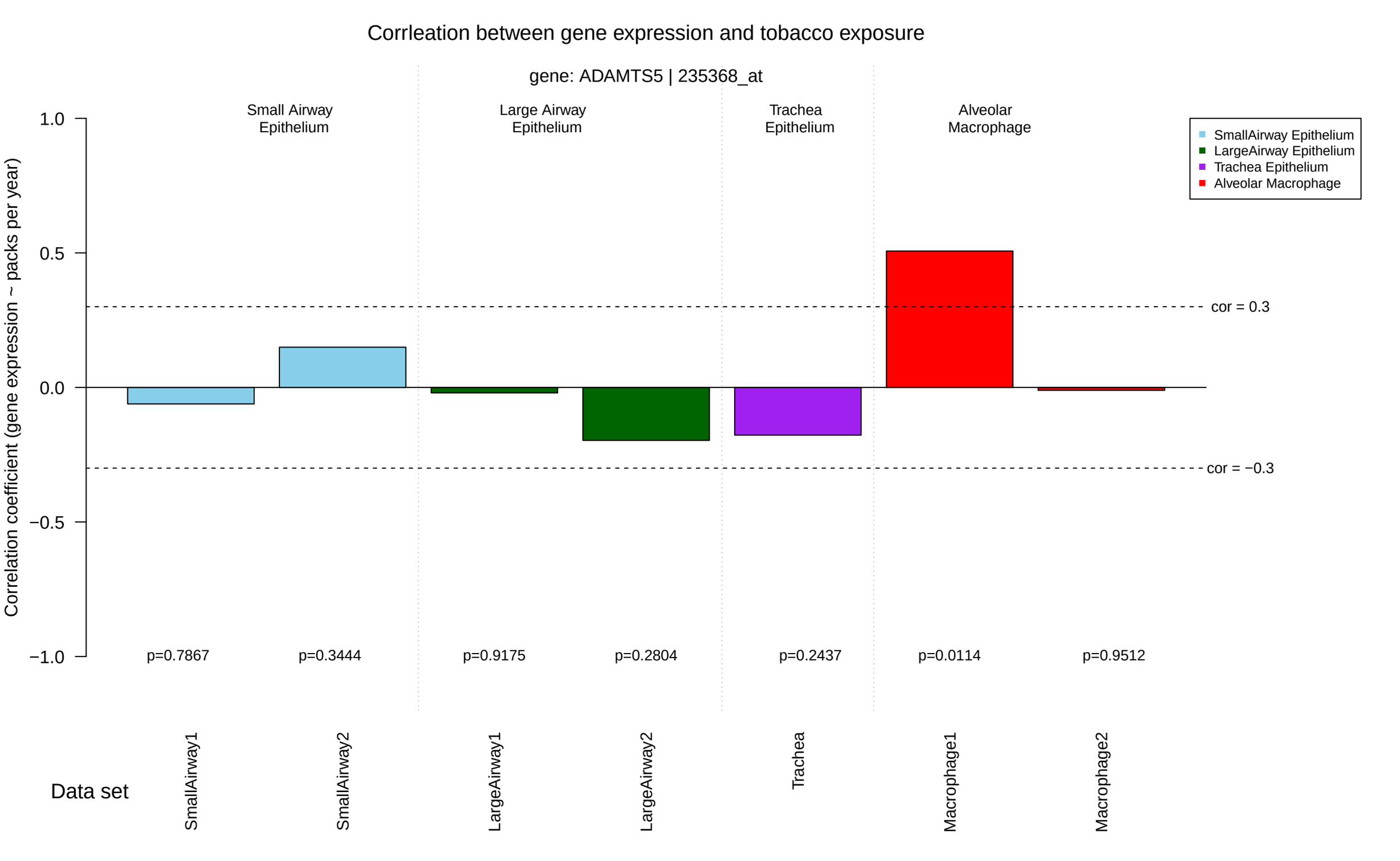

set: LargeAirway1 (p=0.8519)
ADAMTS5 | 235368_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9182)
ADAMTS5 | 235368_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9014)
ADAMTS5 | 235368_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0554)
ADAMTS5 | 235368_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4441)
ADAMTS5 | 235368_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADAMTS5 | 235368_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7867
p=0.3444
p=0.9175
p=0.2804
p=0.2437
p=0.0114
p=0.9512
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2